C Programming Assignment – Question 9

9. Number of Days

Question

Given a year and month number (1 = January through 12 = December), print the number of days in that month. Valid years are 1900–9999. If the year or month is invalid, print 0.

Sample Input

```
2000
2
```

Sample Output

```
29 Days
```

C Program

```c
#include <stdio.h>

int main(void) {
    int year, month, days;

    printf("Enter year and month: ");
    scanf("%d%d", &year, &month);

    if (year < 1900 || year > 9999 ||
        month < 1 || month > 12) {
        printf("0\n");
        return 0;
    }

    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            days = 31;
            break;

        case 4: case 6: case 9: case 11:
            days = 30;
            break;

        case 2:
            if ((year % 400 == 0) ||
                (year % 4 == 0 && year % 100 != 0)) {
                days = 29;
            } else {
                days = 28;
            }
            break;
    }

    printf("%d Days\n", days);
    return 0;
}
```

Explanation

1. Read the year and month.

2. If the year or month is outside the valid range, print 0.

3. Use a switch statement to identify months with 31 or 30 days.

4. For February, check the leap-year rule.

5. Print the number of days.

Note

Year 2000 is a leap year because it is divisible by 400, so February has 29 days.